

# MACHINE LEARNING CASE STUDY REPORT

## Comparative Analysis of Classification Algorithms for Sleep Disorder Prediction

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### Abstract

This case study focuses on evaluating the performance of multiple machine learning classification algorithms for predicting sleep disorder risk using a real-world sleep health dataset. The dataset contains information related to sleep habits, lifestyle, psychological factors, and health conditions of individuals. The target variable was converted into a binary classification problem representing whether a person has a sleep disorder or not.

Three classification algorithms were implemented and compared:

- K-Nearest Neighbors (KNN)
- Naive Bayes (NB)
- Support Vector Machine (SVM)

The models were evaluated using various performance metrics including Accuracy, Precision, Recall, F1-score, True Positive Rate (TPR), False Positive Rate (FPR), Confusion Matrix, and ROC-AUC Curve.

Experimental results showed that the SVM model achieved the best overall performance with an accuracy of 84.51% and AUC score of 0.93.

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### Introduction

Machine Learning classification algorithms are widely used in healthcare systems for disease prediction and risk analysis. Sleep disorders can significantly impact physical and mental health, making early prediction important for preventive healthcare.

The objective of this case study is to compare different classification algorithms and evaluate their effectiveness in predicting sleep disorder risk using sleep-related and lifestyle-related features.

The study also demonstrates how different evaluation metrics help in understanding model performance beyond simple accuracy.

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# Objectives

The objectives of this case study are:

1. To preprocess and analyze the sleep health dataset.
  2. To convert the target variable into a binary classification problem.
  3. To implement multiple classification algorithms.
  4. To evaluate models using:
    - Accuracy
    - Precision
    - Recall
    - F1-score
    - True Positive Rate (TPR)
    - False Positive Rate (FPR)
    - Confusion Matrix
    - ROC Curve
    - AUC Score
  5. To compare model performances and identify the best-performing classifier.
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# Dataset Description

The dataset used in this case study is the Sleep Health Dataset.

## Dataset Details

| Property       | Value                |
|----------------|----------------------|
| Dataset Name   | Sleep Health Dataset |
| Total Rows     | 100,000              |
| Total Columns  | 32                   |
| Missing Values | 0                    |
| Duplicate Rows | 0                    |

The dataset contains information about:

- Demographics
- Sleep patterns
- Lifestyle habits
- Psychological conditions
- Health indicators

- Environmental conditions
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## Target Variable

The original target column:

sleep\_disorder\_risk

contained four categories:

- Healthy
- Mild
- Moderate
- Severe

To simplify the classification problem and enable effective ROC and TPR/FPR analysis, the target was converted into binary classes:

| Original Class | Binary Value |
|----------------|--------------|
| Healthy        | 0            |
| Mild           | 1            |
| Moderate       | 1            |
| Severe         | 1            |

Meaning:

- 0 → No Sleep Disorder
  - 1 → Sleep Disorder Present
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## Feature Selection

Only numerical features were selected to avoid additional preprocessing complexity caused by categorical encoding.

### Selected Features

- age
- bmi
- sleep\_duration\_hrs
- sleep\_quality\_score
- rem\_percentage
- deep\_sleep\_percentage
- sleep\_latency\_mins

- wake\_episodes\_per\_night
  - caffeine\_mg\_before\_bed
  - alcohol\_units\_before\_bed
  - screen\_time\_before\_bed\_mins
  - exercise\_day
  - steps\_that\_day
  - nap\_duration\_mins
  - stress\_score
  - work\_hours\_that\_day
  - heart\_rate\_resting\_bpm
  - sleep\_aid\_used
  - shift\_work
  - room\_temperature\_celsius
  - weekend\_sleep\_diff\_hrs
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## Data Preprocessing

The following preprocessing steps were performed:

1. Dataset loading using Pandas
2. Missing value verification
3. Binary target conversion
4. Feature selection
5. Train-test splitting
6. Feature scaling using StandardScaler

## Train-Test Split

- Training Data: 70%
- Testing Data: 30%

## Feature Scaling

StandardScaler was used to normalize feature values because KNN and SVM algorithms are sensitive to feature magnitudes.

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# Algorithms Used

## 1. K-Nearest Neighbors (KNN)

KNN is a distance-based classification algorithm that predicts classes based on the nearest neighboring data points.

## 2. Naive Bayes

Naive Bayes is a probabilistic classification algorithm based on Bayes theorem with an assumption of feature independence.

## 3. Support Vector Machine (SVM)

SVM is a powerful classification algorithm that finds an optimal decision boundary between classes.

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# Evaluation Metrics

The following metrics were used to evaluate model performance:

## Accuracy

Measures overall prediction correctness.

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

## Precision

Measures how many predicted positive cases are actually positive.

$$\text{Precision} = TP / (TP + FP)$$

## Recall / True Positive Rate (TPR)

Measures how many actual positive cases are correctly identified.

$$\text{Recall} = TP / (TP + FN)$$

## F1-Score

Harmonic mean of Precision and Recall.

$$\text{F1-score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

## False Positive Rate (FPR)

Measures how many negative cases are incorrectly classified as positive.

$$\text{FPR} = \text{FP} / (\text{FP} + \text{TN})$$

## Confusion Matrix

Represents:

- True Positives (TP)
- True Negatives (TN)
- False Positives (FP)
- False Negatives (FN)

## ROC Curve

ROC Curve plots TPR against FPR.

## AUC Score

Area Under the ROC Curve indicating model discrimination capability.

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# Experimental Results

## KNN Results

| Metric       | Value  |
|--------------|--------|
| Accuracy     | 80.18% |
| Precision    | 81.42% |
| Recall / TPR | 73.48% |
| F1-score     | 77.25% |
| FPR          | 14.15% |
| AUC          | 0.86   |

## Confusion Matrix

[[13964 2302] [3642 10092]]

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## Naive Bayes Results

| Metric       | Value  |
|--------------|--------|
| Accuracy     | 82.61% |
| Precision    | 82.62% |
| Recall / TPR | 78.55% |
| F1-score     | 80.53% |

| Metric | Value  |
|--------|--------|
| FPR    | 13.94% |
| AUC    | 0.90   |

### Confusion Matrix

[[13997 2269] [2946 10788]]

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## SVM Results

| Metric       | Value  |
|--------------|--------|
| Accuracy     | 84.51% |
| Precision    | 83.45% |
| Recall / TPR | 82.53% |
| F1-score     | 82.99% |
| FPR          | 13.81% |
| AUC          | 0.93   |

### Confusion Matrix

[[14019 2247] [2399 11335]]

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## Comparative Analysis

| Model       | Accuracy | Precision | Recall/TPR | F1-Score | AUC  |
|-------------|----------|-----------|------------|----------|------|
| KNN         | 80.18%   | 81.42%    | 73.48%     | 77.25%   | 0.86 |
| Naive Bayes | 82.61%   | 82.62%    | 78.55%     | 80.53%   | 0.90 |
| SVM         | 84.51%   | 83.45%    | 82.53%     | 82.99%   | 0.93 |

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## Observations

1. KNN produced good results but had lower recall compared to other models.
  2. Naive Bayes improved classification performance significantly.
  3. SVM achieved the highest performance across almost all metrics.
  4. SVM also produced the highest AUC score, indicating superior class separation capability.
  5. The number of false negatives was lowest in SVM, meaning it missed fewer disorder cases.
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## Advantages of the Proposed Approach

- Real-world healthcare dataset used
  - Multiple models compared
  - Comprehensive evaluation metrics applied
  - ROC-AUC analysis included
  - Effective preprocessing pipeline implemented
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## Limitations

- Only numerical features were used.
  - Binary conversion simplified the original multiclass problem.
  - Deep learning models were not explored.
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## Future Scope

Future improvements may include:

- Multiclass sleep disorder prediction
  - Feature engineering
  - Deep learning approaches
  - Hyperparameter tuning
  - Inclusion of categorical features using encoding techniques
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## Conclusion

This case study successfully implemented and compared KNN, Naive Bayes, and SVM classification algorithms on a sleep health dataset.

The models were evaluated using Accuracy, Precision, Recall, F1-score, TPR, FPR, Confusion Matrix, ROC Curve, and AUC score.

Among all models, SVM achieved the best performance with:

- Accuracy = 84.51%
- Recall = 82.53%
- AUC = 0.93

The experiment demonstrated that SVM is highly effective for predicting sleep disorder risk using sleep and lifestyle-related features.



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## References

1. Scikit-learn Documentation
2. Pandas Documentation
3. NumPy Documentation
4. Machine Learning Concepts by Tom Mitchell
5. Introduction to Statistical Learning
6. Sleep Health Dataset Source